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Comparison of two thermal spike models for ion–solid interaction

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ABSTRACT

A comparative review of the inelastic thermal spike model (ITSM, Meftah et al., 1994) and the analytical thermal spike model (ATSM Szenes, 1995) is given. The ITSM follows the formation of the ion-induced thermal spike based on the Fourier equation while the ATSM skips this stage and a final Gaussian temperature distribution is assumed. Each of the two models doubts the basic assumptions of the other. The ITSM rejects the Gaussian temperature distribution while according to ATSM several thermophysical parameters used by the ITSM are irrelevant to the formation of the thermal spike and the equilibrium values are not valid under spike conditions. The essentially different conclusions of the models are discussed in connection with experiments performed in $BaFe_{12}O_{19}$, Al_2O_3, silica and high-T_c superconductors.

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1. Introduction

There are two thermal spike models, which are often applied in the analyses of the interaction of swift heavy ions with solids: the inelastic thermal spike model (ITSM) (see last review [1]) and the analytical thermal spike model (ATSM) [2]. It is not widely known how significant the differences between them are. A recent paper of Chettah et al. [3] may even increase the confusion regarding the two models as it claims that the ATSM used by Szenes et al. and Dhamodaran et al. coincides with the ITSM saying "our description (ITSM) will complete the recent application of the inelastic thermal spike model by Szenes [4], Dhamodaran [5] et al. and by Kamarou et al. [6] to semiconductors...". The significance of this misstatement to a large extent is due to the co-authors: A. Meftah and M. Toulemonde (corresponding author), who are authors of the present ITSM and its several earlier versions as well. As views of experienced researchers may cause a considerable and prolonged effect on the readers, it was decided to discuss this problem in more details. Our opinion is that because of the widespread application of these two models, it is important to clarify and compare their characteristic features. Such a comparative review of the two thermal spike models will be useful for those who are interested in the experimental and theoretical investigations of the interaction of swift heavy ions with solids.

2. Discussion

The most important features of the two models are summarized in two lists. Some minor points were not included. For the sake of transparency, the comparable features of the two models have identical running numbers in the two lists below. This comparative review may help to understand the advantages and limitations of the models. We emphasize, however, that this is not a critical analysis of the models.

2.1. The ITSM model

A swift heavy ion induces high ionization along its trajectory in a cylindrical region of a solid. In this process, highly charged lattice ions and primary electrons are formed which excite many secondary electrons along their paths. If the charge neutralization time is sufficiently long in the solid the electric interaction between the charged ions leads to an increase of their kinetic energies and a simultaneous reduction of the electric energy density. It is assumed that the strong repulsion forces between lattice ions may destroy the local structure if they exceed the mechanical strength of the solid. This process is related to the primary stage of ionization and it is the physical basis of the Coulomb explosion theories which were applied previously for the explanation of various ion-induced effects [7–9].

A general problem of various theories of ion–solid interaction that they require values of some macroscopic materials parameters in a physical state when the lattice atoms are strongly ionized by the projectile. However, the ionization density is so high in the primary stage that even the physical meaning is uncertain for the

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materials parameters (Young's modulus and dielectric constant in the case of the Coulomb-explosion mechanism).

After the charge neutralization is finished in a solid, the dominant fraction of the deposited energy is in the electron system. During the relaxation process this energy is transferred to the lattice atoms, leading to the formation of a high temperature region called the thermal spike.

In the ITSM [1], the energy transport and the ion-induced damage formation is described by the Two-Temperature Model (TTM) [10]. In the TTM, local equilibrium is assumed in the electron and phonon systems which are characterized by the electron temperature T_e and lattice temperature T_a. The heat flow in the lattice (a) and the electron (e) system and between them is described by two coupled Fourier equations;

$$C_e T_e \frac{\partial T_e}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left[rK_e(T_e)\frac{\partial T_e}{\partial r}\right] - G(T_e - T_a) + A(r,t) \tag{1a}$$

$$C_a T_a \frac{\partial T_a}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left[rK_a(T_a)\frac{\partial T_a}{\partial r}\right] + G(T_e - T_a). \tag{1b}$$

Here C, K and T are the volume-specific heat capacity, thermal conductivity and temperature, respectively, r is the radial distance from the trajectory and t is the time. The coupling between the systems is due to the electron–phonon interaction term $G(T_e - T_a)$. A mean value of the electron–phonon coupling factor G is used in a broad range of electron energies. $A(r, t)$ is the energy density per unit time on the target electrons. The actual form of $A(r, t)$ is based on the radial energy distribution of secondary electrons given by Waligorski [11]. The parameters of the electron system including the carrier densities are calculated applying the free electron model and assuming that they do not depend on r and t.

The most important features of the ITSM are the following (see [1] if references are missing)

i. The heat diffusion in the electron and lattice subsystems, the process of the formation of the thermal spike are described by the numerical solution of coupled Fourier equations assuming local equilibrium and an explicit functional form of the deposited energy density $A(r, t)$.
ii. It is assumed that the relaxation of the excited electron system proceeds exclusively by electron–phonon coupling starting from $t < 1$ fs.
iii. The thermal energy of the spike ε increases up to the electronic stopping power S_e.
iv. The track size is defined by the maximum radial zone which contains sufficient energy for melting (defined by the energy to reach the melting temperature T_m plus the heat of fusion L).
v. The model provides a non-analytical R_e–S_e (R_e – effective track radius) relationship which is valid for cylindrical continuous tracks ($R_e > 2$ nm [12], $R_e > 3$ nm [13] for α-SiO_2 and $Y_3Fe_5O_{12}$, respectively).
vi. The electron–phonon mean free path λ derived from the electron–phonon coupling constant G, and the heat of fusion L (when its value is not known) are the free parameters of the model.
vii. The velocity effect (larger tracks are induced at low than high ion velocities with S_e = constant) is attributed to the spatial broadening of the initial energy deposition with increasing specific ion energy E [14].
viii. Specific features.
   (a) An empirical relationship is found between λ and the band gap energy E_g.
   (b) The equilibrium values are used for the thermophysical parameters (thermal conductivity, specific heat, the heat of fusion L, heat of evaporation and temperature coefficients) within the transient conditions of the spike.

According to the authors, "drastic approximations" characterize the ITSM [13]. In the above incomplete list i, ii, iii, iv, vi, vii and viii(b) are various assumptions of the model. The model has been applied to various ion-induced effects in insulators, semiconductors and metals and to materials with highly anisotropic conduction as well.

2.2. The ATSM model

In the ATSM it is realized that the actual time evolution of the spike temperature up to the maximum is unimportant for various ion-induced physical effects. It is assumed that the ion-induced temperature increase $\Delta T(r,t)$ can be approximated by a Gaussian distribution function and at the moment of maximum peak temperature T_p ($t = 0$), it is described by

$$\Delta T(r,0) = \frac{gS_e}{\pi\rho c a^2(0)} \exp -\{r^2/a^2(0)\}, \tag{2}$$

where ρ and c are the density and the specific heat given by the Neumann–Kopp's rule, and the two parameters of the model are the efficiency g and the initial Gaussian width $a(0)$.

The R_e effective radius of the tracks can be found from the condition

$$\Delta T(R_e,0) = T_o = T_m - T_{ir}, \tag{3}$$

where T_{ir} is the irradiation temperature. Under these assumptions, three simple equations can be derived from Eq. (2)

$$R_e^2 = a^2(0)\ln(S_e/S_{et}), \quad \text{for } S_e < 2.7S_{et}, \tag{4}$$

$$R_e^2 = \frac{a^2(0)S_e}{2.7S_{et}}, \quad \text{for } S_e > 2.7S_{et}, \tag{5}$$

$$S_{et} = \frac{\pi\rho c T_o a^2(0)}{g}, \tag{6}$$

where S_{et} is the threshold value of S_e for track formation. By introducing Eq. (6) into Eq. (5), the parameters $a(0)$ and the g can be obtained separately in the analysis.

The most important features of the ATSM are the following (see [2] if the reference is missing)

i. The ion-induced temperature increase in the lattice $\Delta T(r, t)$ is approximated by a Gaussian function for $t \geqslant 0$; the peak temperature is maximum at $t = 0$.
ii. The mechanisms of the energy deposition into the lattice are not specified.
iii. Only a fraction of the deposited inelastic energy gS_e appears as the thermal energy of the spike ε.
iv. The track size is proportional to the maximum volume of the melt defined by the maximum radial zone with $T = T_m$ (L is neglected, see later). The proportionality factor is $k = 1$ for those solids for which scaling (see viii) is valid.
v. Analytical equations are obtained for the threshold for track formation S_{et} and for the track evolution R_e^2–S_e.
vi. The initial ($t = 0$) Gaussian width $a^2(0) = d\ln R_e^2/dS_e$ ($R_e < a(0)$) and the efficiency g are the two parameters of the model.
vii. The origin of the velocity effect is the change of the thermal energy $\varepsilon = gS_e$ with the specific ion energy E [15].
viii. Specific features in track-forming insulators.
   (a) A relation $S_{et}/(\rho c(T_m - T_{ir}))$ = constant is confirmed by experiments at low ion velocities ($E < 2$ MeV/nucleon) and high ion velocities ($E > 8$ MeV/nucleon) as well, which demonstrates that the track formation proceeds through local melting [2,16].

(b) The $R_e^2 - S_e/(\rho c(T_m - T_{ir}))$ normalized track evolution curves coincide in various solids at low and high ion velocities as well. The absence of L from the scaling factor is due to its considerable reduction below the equilibrium value under spike conditions. This master curve demonstrates that the initial Gaussian width is uniformly $a(0) = 4.5$ nm, and $g \approx 0.4/0.17$ at low/high ion velocities. Thus there are no individual (different for various solids) free parameters for insulators at low and high ion velocities in this model [2,17].

In the above list, i, iv, vi and viii(a) (Neumann–Kopp's rule) are the assumptions of the model, ii, iii, v, vii, viii(a) (except the validity of the Neumann–Kopp's rule), viii(b) are various conclusions obtained from the analysis of the experimental data. The ATSM has been applied to insulators, semiconductors, polymers, to biological samples and to materials with highly anisotropic conduction as well.

2.3. Advantages, assumptions and contradictions in the two models

As it is obvious from the above brief review, the two models are dissimilar: each contradicts the basic assumptions of the other. The temperature distribution derived in the ITSM is non-Gaussian. Moreover, ε may be up to 15 times higher than in the ATSM for the same irradiation experiment [18] while the maximum cross-sectional areas at $T = T_m$ are equal (iv) to ensure the agreement with the experimental track sizes. On the other hand, it is obvious that basic parameters of the track formation process cannot be missing from the scaling factor $\rho c(T_m - T_{ir})$ in viii(b) of ATSM. Thus the successful scaling (master curve) is in conflict with the ITSM since it implies that the various thermal parameters applied in ITSM (thermal conductivity, heat of fusion, heat of evaporation, heat of sublimation, various temperature coefficients) are indifferent for the track diameter. Additionally, viii(b) for ATSM demonstrates that the equilibrium value of the heat of fusion L is not valid under the conditions of the thermal spike in contradiction with the ITSM. The study of the sputtering [19] and ion beam mixing [20] led to similar conclusions for other thermal parameters. The explanation of the physical origin of the so-called velocity effect (vii) is also diametrically opposite.

The ATSM is not committed to any mechanism of the energy deposition to the lattice. Thus it can incorporate without any problem various contributions to the thermal energy. For example, in the case of the Coulomb-explosion mechanism the energy of the Coulomb field is transformed partially to the kinetic energy of lattice ions and this increases ε [21–23]. This is not true for the ITSM which assumes the electron–phonon coupling exclusively (ii).

The advantage of the ITSM model is that it provides the reader with the impression that the basic mechanism of the energy transfer has been uncovered. The model can be easily applied by anyone since the authors provide a free access to their code.

The advantage of the ATSM is its simplicity, transparency and its high predictive power. However, the model does not explain why only a small fraction of S_e appears as the thermal energy of the spike.

The various assumptions of the ITSM have not been verified up until now. The only evidence is the agreement between the measured and calculated track radii achieved by two fitting parameters λ and L. In the ATSM i is justified by the experiments (see Section 2.4) iv is confirmed by correlation given in viii(a) and by the fact that k = constant in iv for various solids according to the experiments. Additionally, it is evident that $k = 1$ is the only reasonable value [2]. The validity of the Neumann–Kopp's rule for the specific heat is a further assumption which is justified indirectly by the two correlations in viii. Up till now the consequences of significant deviations from the Neumann–Kopp's rule have not been observed in these two correlations.

In ITSM and ATSM T_m measured under normal conditions is used. However, the laws of thermodynamics predict superheating for very rapid changes of temperatures. Thus superheating is expected under spike conditions as well [24]. Reliable estimates of the superheating temperatures can be simply introduced into the two models.

Recently, Klaumünzer expressed doubts concerning the application of the classical heat diffusion equation under the conditions of an ion-induced spike [24]. Indeed, the Fourier equation leads to incorrect results when the mean free path of the heat carriers (e.g. $\lambda \approx 4$–13 nm [1]) is comparable to or larger than the linear dimension of the object of interest [25] (excited region, e.g. $R_e = 1$–5 nm). This is a serious problem for ITSM which is based on the numerical solution of the coupled Fourier equations (i). Although the criticism of Klaumünzer impugns the physical basis of the ITSM, no appropriate response has been published up until now. The reliability of the predicted spike temperatures is uncertain at present, however, this may be changed after the response of the authors of ITSM.

The criticism does not concern the ATSM since the Gaussian temperature distribution in the ATSM is not related to the Fourier equation. Mozumder considered the Gaussian distribution as the result of the random nature of the process of thermalization of the excited electrons [26]. In the ATSM the Gaussian temperature distribution is an assumption of the model whose validity has been justified experimentally.

2.4. Experimental information about the spike parameters

The temperature distribution in the spike is the most important information for both thermal spike models. The spike temperatures cannot be measured directly. However, indirect information can be obtained by making use of the dependence of the track size on the parameters of the temperature distribution in the spike. While there is no such possibility for ITSM, Eqs. (4)–(6) provide analytical relationship between the track parameters and the parameters of the temperature distribution in ATSM.

The validity of Eqs. (4) and (5) derived from the assumption Eq. (2) has been justified for a number of track-forming insulators in broad ranges of S_e and E [17,27]. It has been also shown that there is a transition from the logarithmic (Eq. (4)) to a linear (Eq. (5)) track evolution at about $S_e \approx 2.7S_{et}$ as predicted [29], and the values of S_{et} are close when derived either from Eq. (4) or (5) [27].

Additionally, the initial Gaussian width of the distribution $a(0)$ has the same value in track-forming insulators in the range 0.02–20 MeV/nucleon [30] and there are no free parameters according to viii(b). The demonstration of these uniform features for insulators by ATSM would be impossible if the difference between the real and the assumed $\Delta T(r, t)$ distributions were significant. By using the above uniformity, Eq. (6) has been also proven by using systematic experiments on a number of solids [2,16,17,31]. Finally, an $R_e^2 - S_e/(\rho c(T_m - T_{ir}))$ master curve is derived for insulators [viii(b)] and is verified by the experimental data [17,27].

Each of these features directly or indirectly is the consequence of Eq. (2). The master curve is a very strong argument in favour of the Gaussian distribution as it is a relationship between raw experimental data without any fitting parameters. Thus the quantitative agreement with the predictions and the uniform behaviour of track-forming insulators revealed by the ATSM proves its basic assumption: the formation of an ion-induced Gaussian temperature distribution.

When the temperature profiles predicted by the two models were compared, Klaumünzer concluded that the differences between the two temperatures disappear in the 10–100 ps range.

However, the agreement between the temperature profiles is impossible at any time as the thermal energies may differ by an order of magnitude [18]. The origin of this large difference is related to the features of the models under iii.

The inelastic energy deposition induces various processes which contribute to the energy balance. Elastic waves can carry off alone up to 50% of the initially deposited energy of a spherical spike [32] or up to 30% for a cylindrical spike [33]. Besides elastic waves defect formation, deformation and radiative processes also may lead to a considerable decrease of the energy available for ε. Nevertheless, it is an assumption in the ITSM that $\varepsilon/S_e = 1$ i.e. S_e is transferred to thermal energy without any loss. In the ATSM $\varepsilon/S_e = g$ and the value of the efficiency g is determined by fitting Eqs. (4)–(6) to the experimental data. The results show that g is not an individual fitting parameter, which has different values for various solids. The efficiency g has very close values for track-forming insulators: $g = 0.4$ for low ion velocities ($E < 1$–2 MeV/nucleon) and $g = 0.17$ for high ion velocities ($E > 8$ MeV/nucleon) [30]. Thus $g < 1$ is not an additional assumption of the model but rather a conclusion from the analyses. The reduction of g leads to an increase of S_{et} (see Eq. (6)) and smaller tracks are induced in this case (see Eqs. (4), (5)). This is the origin of the velocity effect.

The fact that g decreases with increasing E is an indication of an ion velocity-dependent contribution to ε. The negligible velocity effect in high-T_c superconductors shows that the time of neutralization of lattice ions by electrons may have an important role in this contribution [18]. However, this has not been discussed theoretically yet.

We note that in ATSM ε is the thermal energy for a Gaussian temperature distribution when T_p = maximum ($t = 0$) is attained. Contributions to the thermal energy in later moments during cooling or outside the Gaussian spike do not affect the track size, therefore, they are not included into ε of the ATSM. A systematic error due to superheating may also modify the estimated value of the efficiency g (only the ratio $(T_m - T_{ir})/g$ can be estimated from the experiments).

2.5. The physical origin of the differences between ITSM and ATSM

The fundamental physical idea is the same in both models: the formation of an ion-induced thermal spike. In principle, the two models might lead to similar results. However, the calculations of the spike parameters are made along different physical assumptions in the two models.

In the ITSM the formation of the thermal spike is followed in details. Fourier heat equation and equilibrium values of the thermophysical parameters [viii(b)] are applied, which require local equilibrium. This is the application of a conventional theoretical method as if the spike conditions were a typical and not an extraordinary problem of the heat transport. However, the mean free path of the heat carriers is too long (see the last paragraphs of Section 2.3), changing the mechanism of the heat transfer from diffusive to ballistic and breaking the local equilibrium condition. This is a serious limitation of the validity of the Fourier equation, which assumes a diffusive process.

The ATSM does not rely on the Fourier equation. The formation process of the thermal spike is skipped, therefore, the actual mechanism of the heat transport loses its importance. The Gaussian functional form of the spike temperature is introduced as a separate assumption avoiding its derivation from the heat equation (the Fourier equation has a Gaussian solution). The equilibrium values of the various thermophysical materials parameters are rejected (except for T_m) and their values under spike conditions are derived from irradiation experiments.

To perform a calculation according to ITSM the knowledge of the following parameters is required: T_m, T_v (vaporization temperature), $C_a(T)$, $K_a(T)$ for $T < T_v$, ρ in the solid and liquid phases, L and latent heat of vaporization, C_e and D_e [3]. In contrast to this, an uniform behaviour is revealed by using the ATSM. If N is the number of atoms per unit volume, the equation $\rho c = 3Nk$ (k – Boltzmann constant) holds and Eq.(2) is simplified to the form

$$\Delta T(r,0) = \frac{g < s_e >}{3\pi k a^2(0)} \exp - \{r^2/a^2(0)\}, \tag{7}$$

where $<s_e> = S_e/N$. In Eq. (7), $a(0) = 4.5$ nm for all track-forming insulators and the efficiency g has similar values for these solids in a broad range of E. Thus, at equal values of $<s_e>$ the same $\Delta T(r,0)$ is induced in various track-forming insulators. When Eq. (7) is applied to Eq. (3), we obtain

$$T_m - T_{ir} = \frac{g < s_e >}{3\pi k a^2(0)} exp - \left\{R_e^2/a^2(0)\right\}. \tag{8}$$

According to Eq. (8) the track size depends only on the single materials parameter T_m. When tracks are measured in n different solids at $<s_e>$ = constant, and the data pairs T_{mi}, R_{ei}^2, ($i = 1 \ldots n$) are plotted, then Eq. (8) predicts that the independent data measured in various solids form a Gaussian curve with width $a(0)$ in a $(T_m - T_{ir}) - R_e^2$ plot. Such a plot has been made by using published experimental data, and good quantitative agreement with Eq. (8) was obtained [28]. We emphasize that the plot demonstrates a relationship between the raw experimental data without any adjustable parameters. Consequently, the Gaussian curve in the T_m–R_e^2 plot reveals an intrinsic property of the phenomenon.

Thus the main difference between ITSM and ATSM is in the characterization of the transient spike conditions. In the ITSM it is assumed that the spike temperature is formed in a very complex process which depends sensitively on many materials parameters. ITSM assumes local equilibrium and applies a method suitable only for the diffusive heat transport.

The ATSM denies local equilibrium in the spike and uses a simplification which does not require the knowledge of the mechanism of the heat transport. It is established and verified experimentally, that $\Delta T(r,0)$ is a Gaussian function, which is identical in track-forming insulators for $<s_e>$ = constant, and it does not depend on any of the materials parameter listed above for ITSM.

2.6. Some divergent results

Some principal differences have been already discussed in previous sections. In this section we discuss some practical aspects.

Most of the irradiation experiments have been performed for the study of the formation of amorphous tracks. In general, the two models reproduce the track sizes with comparable accuracy. According to the ATSM, the track-forming insulators exhibit identical behaviour in normalized plots (viii.). Thus, some large deviations from the $R_e^2 - S_e/(\rho c(T_m - T_{ir}))$ master curve are easily observed, and they can be taken into account with a lower weight in an extrapolation or are neglected as an experimental error. This is especially important for small track sizes as they may affect significantly the estimate of S_{et}.

The deviations are masked by the fitting procedure in ITSM. Therefore, the two models sometimes lead to considerably different values of S_{et}. Additionally, the ITSM is valid for relatively large tracks. The minimum value is $R_e = 3$ nm for $Y_3Fe_5O_{12}$ (v) while $R_e < 6$ nm for the largest tracks measured at high ion velocities [14]. Besides the above reasoning, the extrapolation to $R_e = 0$ of an R_e–S_e curve (ITSM) and according to Eq. (4) (ATSM) evidently leads to different results. A typical example is α-SiO_2 for which

$S_{et} \approx 1{,}5$ keV/nm [34] and $S_{et} \approx 4.5$ keV/nm [29] were derived from the same data.

The R_e^2–S_e track evolution cannot be described by the equations of the ATSM in Al_2O_3 and $MgAl_2O_4$, as the tracks are much smaller than predicted. However, the experiments show that viii(a) is valid for them as well. Thus track formation proceeds through melting as in other insulators. It was concluded that the observed deviations from the predictions are due to the fast partial crystallization of the amorphous tracks [27]. The ITSM suggests a rather sophisticated model and the change of the criterion iv for sapphire requiring vaporization of the material instead of melting [35].

Another example of the different interpretation is the electronic sputtering in SiO_2 which is a thermally activated process in both models. The quasi-equilibrium value of the sublimation energy (5.4 eV) was used for the activation energy U in the ITSM [35,36] and only $U = 0.42$ eV was deduced with the ATSM from a larger set of experimental data [19] including those in Refs. [35,36]. This large deviation of the activation energies demonstrates the large difference between the two approaches. A sputtering threshold $S_e \approx 10$ keV/nm was estimated in Ref. [36] by Toulemonde et al. while the threshold was not observed in Ref. [19]. Otherwise, a threshold value is not consistent with the thermal activation mechanism as the sputtering has merely a low probability at low values of S_e without a threshold.

There are also considerable differences in the interpretation of the experimental results on non-insulating solids with highly anisotropic conduction. Earlier, Zhu et al. [37] demonstrated by transmission electron microscopy that the track cross-sections varied by 50% and 110% when the beam direction was changed in high-T_c superconductors (HTCSs) Y–Ba–Cu–O and Bi–Sr–Ca–Cu–O. Thus track formation is highly anisotropic in these solids. The analysis according to ATSM shows that the $R_e^2 - S_e$ curves follow Eq. (4) with $a(0) \approx 4.5$ nm [18] corresponding to a narrow spike like in insulators. This is an indication that the flux of charge carriers in the direction of low conductivity controls the track formation. The anisotropy is due to the flux of charge carriers along the planes with metallic conduction: their contribution changes with the beam direction [18].

In the ITSM, HTCSs are considered as isotropic metals ($E_g = 0$) in which the thermal spike is broad and a high value of $\lambda = 12.8$ nm is obtained from the analysis [38]. This high value of the electron–phonon mean free path λ is in obvious contradiction with $a(0) \approx 4.5$ nm, as the latter characterizes the insulators with a narrow spike for which $\lambda \approx 4$-5 nm [1]. We note that $a(0)$ is derived directly from the raw experimental data as $a^2(0) = d\ln R_e^2/dS_e$ while λ is a fitting parameter whose reliability depends on the validity of many "drastic approximations" [13]. Actually, the free-electron approximation is applied in the analysis [1,38], which is evidently not suitable for describing the anisotropy of track formation. Similar contradictions can be found in the analyses of the experiments on the highly anisotropic semiconductor GeS published in [38] (ITSM) and [39] (ATSM). Additionally, the tracks are elliptical in GeS and elliptical tracks also occur in HTCSs [37]. However, ITSM is valid only for cylindrical tracks (v).

Track formation in $BaFe_{12}O_{19}$ is characterized by the ITSM with a mean value $\lambda = 8.2$ nm [40] while $a(0) \approx 4.5$ nm for these same tracks [2]. According to the ATSM, the spike is as narrow as in other insulators, while the ITSM indicates a considerably broader spike. The situation is similar to the results in HTCSs discussed in the previous paragraph.

We note that there is a considerable drop in the R_e–S_e curve of $BaFe_{12}O_{19}$ at high values of S_e. There is a corresponding deviation from the $R_e^2 - S_e/(\rho c(T_m - T_{ir}))$ master curve as well. If the deviation is a physical effect, then $BaFe_{12}O_{19}$ is a unique exception, showing a standard behaviour for $R_e < a(0)$ with $a(0) \approx 4.5$ nm and a systematic deviation from the master curve at higher values of S_e. As such behaviour has not been observed yet in other insulators, we are inclined to consider this drop as the consequence of a systematic experimental error.

There was an attempt to explain the drop of track sizes in $BaFe_{12}O_{19}$ by the velocity effect (vii) according to the ITSM. The authors presented two separate theoretical R_e–S_e curves for $E = 7$ MeV/nucleon ($\lambda = 6.9$ nm) and $E = 23$ MeV/nucleon ($\lambda = 9.5$ nm) [40] and reported a good agreement between the experimental results and the theoretical calculations. We note that the electron–phonon mean free path λ does not change with E in $Y_3Fe_5O_{12}$ [1] and the authors often emphasize this feature. The unexplained deviation from this standard behaviour in $BaFe_{12}O_{19}$ throw doubts on the analysis.

Unfortunately, the projectile specific energies were not shown in the figure and were not reported in the paper either. However, the comparison with the original experimental data [41,42] shows that the track radii are not separated in this plot according to their specific energies as predicted but they are equally distributed along both theoretical curves. Thus the drop of the track sizes is not related to the velocity effect and, therefore, the $\lambda = 8.2$ nm deduced for $BaFe_{12}O_{19}$ with this assumption is rather unreliable.

The two models lead to quite contradictory results for highly anisotropic crystals (Y–Ba–Cu–O, GeS) and for $BaFe_{12}O_{19}$ as well. This is a delicate point as the parameters of just these solids prove the relationship between λ and E_g provided by the ITSM [1,35]. Our opinion is that the results presented up until now are not suitable to confirm reliably the existence of a λ–E_g relationship. As semiconductors with high value of E_g (diamond $E_g = 5.5$ eV, AlN $E_g = 6.2$ eV) also occur, the existence of a unique λ–E_g seems to be problematic.

The estimate of S_{et}, the sputtering in SiO_2, the track analysis in sapphire and $BaFe_{12}O_{19}$ show the principal differences between the two approaches. This is not valid for the results on HTCSs and GeS as the preconditions of the ITSM: circular track cross section and isotropy for the electrons are not fulfilled in these solids.

3. Conclusions

We have presented a comparative review of the ITSM and ATSM. The two models are quite different and the basic assumptions and the results of each model contradict to those of the other model. The $R_e^2 - S_e/(\rho c(T_m - T_{ir}))$ master curve in the ATSM throws doubt on the role of various thermal parameters and their equilibrium values (thermal conductivity, specific heat, the heat of fusion L, heat of evaporation) in track formation. In fact, L is negligible according to ATSM while L is a free parameter in ITSM. On the other hand, the validity of the basic assumption of ATSM, the Gaussian temperature distribution is doubted by the ITSM.

The main physical difference between ITSM and ATSM is in the characterization of the transient spike conditions. According to ATSM, $\Delta T(r, 0)$ is a Gaussian function which is identical in track-forming insulators for $\langle s_e \rangle$ = constant, while it depends sensitively on a number of materials parameters in ITSM. The mechanism of the heat transport changes from diffusive to ballistic in a spike induced by swift heavy ions while the ITSM applies the Fourier equation which assumes heat diffusion. The ATSM uses a simplification which does not require the knowledge of the mechanism of the heat transport. The ITSM assumes local equilibrium, the ATSM denies it.

Although both models can explain most of the experimental data, they lead to considerably different conclusions in the analysis of the experiments in $BaFe_{12}O_{19}$, Al_2O_3, silica and HTCSs. The existence of a λ–E_g relationship derived by the ITSM requires further evidences.

References

[1] M. Toulemonde, W. Assmann, C. Dufour, A. Meftah, F. Studer, C. Trautmann, Mat. Fys. Med. 5 (2006) 263.
[2] G. Szenes, Phys. Rev. B 51 (1995) 8026.
[3] A. Chettah, H. Kucal, Z.G. Wang, M. Kac, A. Meftah, M. Toulemonde, Nucl. Instr. Meth. B 267 (2009) 2719.
[4] G. Szenes, Z.E. Horváth, B. Pécz, F. Pászti, L. Tóth, Phys. Rev. B 65 (2002) 045206.
[5] S. Dhamodaran, A.P. Pathak, A. Dunlop, G. Jaskierowicz, S. Della Negra, Nucl. Instr. Meth. B 256 (2007) 229.
[6] A. Kamarou, W. Wesch, E. Wendler, A. Undisz, M. Rettenmayr, Phys. Rev. B 73 (2006) 184107.
[7] R.L. Fleischer, P.B. Price, R.M. Walker, J. Appl. Phys. 36 (1965) 3645.
[8] S. Klaumünzer, M.D. Hou, G. Schumacher, Phys. Rev. Lett. 57 (1986) 850.
[9] D. Lesueur, A. Dunlop, Radiat. Eff. Def. Sol. 126 (1993) 163.
[10] M.I. Kaganov, I.M. Lifshitz, L.V. Tanatarov, Zh. Eksp. Teor. Fiz. 31 (1956) 232.
[11] M.P.R. Waligorski, R.N. Hawn, R. Katz, Nucl. Track. Radiat. Meas. 11 (1986) 309.
[12] A. Meftah, F. Brisard, J.M. Costantini, E. Dooryhee, M. Hage-Ali, M. Hervieu, J.P. Stoquert, F. Studer, M. Toulemonde, Phys. Rev. B 49 (1994) 12457.
[13] A. Meftah, J.M. Costantini, M. Djebara, N. Khalfaoui, J.P. Stoquert, F. Studer, M. Toulemonde, Nucl. Instr. Meth. B 122 (1997) 470.
[14] A. Meftah, F. Brisard, J.M. Costantini, M. Hage-Ali, J.P. Stoquert, F. Studer, M. Toulemonde, Phys. Rev. B 48 (1993) 920.
[15] G. Szenes, Phys. Rev. B 52 (1995) 6154.
[16] G. Szenes, Phys. Rev. B 61 (2000) 14267.
[17] G. Szenes, F. Pászti, Á. Péter, A.I. Popov, Nucl. Instr. Meth. B 166–167 (2000) 949.
[18] G. Szenes, Phys. Rev. B 54 (1996) 12458.
[19] G. Szenes, Nucl. Instr. Meth. B 233 (2005) 70.
[20] G. Szenes, Appl. Phys. Lett. 81 (2002) 4682.
[21] E. Seiberling, J.E. Griffith, T.A. Tombrello, Radiat. Eff. 52 (1980) 201.
[22] S. Klaumünzer, Nucl. Instr. Meth. B 225 (2004) 136.
[23] G. Szenes, Nucl. Instr. Meth. B 146 (1998) 420.
[24] S. Klaumünzer, Mat. Fys. Medd. Dan. Vid. Selsk 52 (2006) 293.
[25] M.I. Flik, B.I. Choi, K.E. Goodson, J. Heat Transfer 114 (1992) 666.
[26] A. Mozumder, Adv. Rad. Chem. 1 (1969) 1.
[27] G. Szenes, J. Nucl. Mat. 336 (2005) 81.
[28] G. Szenes, Radiat. Eff. Def. Sol. 162 (2007) 557.
[29] G. Szenes, F. Pászti, Á. Péter, D. Fink, Nucl. Instr. Meth. B 191 (2002) 186.
[30] G. Szenes, D. Fink, S. Klaumünzer, F. Pászti, Á. Péter, Nucl. Instr. Meth. B 245 (2006) 243.
[31] V. Martynenko Yu, M.V. Umansky, Radiat. Eff. Def. Sol. 132 (1994) 31.
[32] M.M. Jakas, E.M. Bringa, Phys. Rev. B 62 (2000) 824.
[33] G. Szenes, Phys. Rev. B 60 (1999) 3140.
[34] N. Khalfaoui, C.C. Rotaru, S. Bouffard, E. Jacquet, H. Lebius, M. Toulemonde, Nucl. Instr. Meth. B 209 (2003) 165.
[35] M. Toulemonde, Ch. Dufour, A. Meftah, E. Paumier, Nucl. Instr. Meth. B 166–167 (2000) 903.
[36] M. Toulemonde, W. Assmann, C. Trautmann, F. Grüner, Phys. Rev. Lett. 88 (2002) 057602.
[37] Yimei Zhu, Z.X. Cai, R.C. Budhani, M. Suenaga, D.O. Welch, Phys. Rev. B 48 (1993) 6436.
[38] A. Meftah, J.M. Costantini, N. Khalfaoui, S. Boudjadar, J.P. Stoquert, F. Studer, M. Toulemonde, Nucl. Instr. Meth. B 237 (2005) 563.
[39] G. Szenes, Radiat. Eff. Def. Solids 161 (2006) 401.
[40] M. Toulemonde, E. Paumier, J.M. Costantini, C. Dufour, A. Meftah, F. Studer, Nucl. Instr. Meth. B 116 (1996) 37.
[41] A. Meftah, N. Merrien, N. Nguyen, F. Studer, H. Pascard, M. Toulemonde, Nucl. Instr. Meth. B 59–60 (1991) 605.
[42] F. Studer, C. Houpert, H. Pascard, R. Spohr, J. Vetter, Y.F. Jin, M. Toulemonde, Radiat. Eff. Def. Solids 116 (1991) 59.